

Applied AI Workshop

Welcome!

Today's Agenda

Wednesday July 22		
Time	Activity	Lead
1:00 pm - 2:30 pm	Hands-On: Designing of ML Models	Edgar Lobaton
2:30 pm - 3:00 pm	Optimization and Regularization	Edgar Lobaton
3:00 pm - 4:00 pm	Advanced Architectures: Transformers and Diffusion Models	Edgar Lobaton
4:00 pm - 4:30 pm	From Generative Models to AI Assistants	Edgar Lobaton
4:30 pm - 5:00 pm	Open Discussion	Edgar Lobaton

ML Development Workflow Example

Edgar Lobaton, Ph.D.

Director of Applied AI, College of Engineering (COE), NC State
Professor, Electrical and Computer (ECE) Engineering, NC State
NAIRR AI Education Fellow, Computing Research Association (CRA)

ML Workshop Example

Problem at Hand: Counting and Trait Extraction of Blueberries

Steps:

- Understanding the Data and Problem
- Formalizing the Tasks
- Data Preparation and Annotation:
 - Selecting a Schema
 - Hierarchical Annotation
 - AI Assisted Annotation
- Developing and Evaluating the Model
- Deploying the Model



AI Collaboration Modes for Content Generation

Edgar Lobaton, Ph.D.

Director of Applied AI, College of Engineering (COE), NC State
Professor, Electrical and Computer (ECE) Engineering, NC State
NAIRR AI Education Fellow, Computing Research Association (CRA)

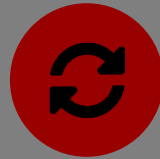
Three Modes of Working with an AI Code Assistant

Cyborgs, Centaurs, and Self-Automators — from a Harvard Business School study of AI-assisted knowledge work



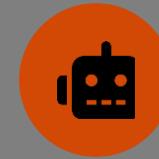
Centaur

A clean division of labor. You decide what needs to happen, then delegate one bounded piece to the AI — and take control back to verify it.



Cyborg

Continuous interleaving, not a clean split. You and the AI go back and forth repeatedly within the same task — no clear line between “your part” and “its part.”



Self-Automator

The whole task goes to the AI. One high-level prompt, minimal ongoing engagement, review only at the end.

Let's think of examples of use on non-coding tasks

Same ML Task, Three Different Workflows

What each mode looks like when building a model with an AI code assistant

	Centaur	Cyborg	Self-Automator
Your role	<i>Decide architecture & strategy yourself</i>	<i>Co-write live — start, edit, hand off, repeat</i>	<i>Write one high-level prompt</i>
AI’s role	<i>Handles one bounded, delegated task</i>	<i>Continuously drafts, edits, explains errors with you</i>	<i>Runs the pipeline end-to-end, largely unattended</i>
Example prompt	<i>“Write the training loop for this architecture”</i>	<i>Live back-and-forth fixing a failing test together</i>	<i>“Build a model that predicts X from this dataset”</i>

None of these is “the right way” — the point is recognizing which mode you’re in and choosing it on purpose.

Designing and ML Workflow

A photograph of a modern, multi-story building at dusk. The building features a large glass facade on the right side, which reflects the vibrant orange and pink colors of the sunset. The rest of the building has a dark, textured facade with horizontal slats. Several windows are illuminated from within, showing a warm yellow light. The sky is filled with dramatic, colorful clouds in shades of blue, purple, and orange. In the foreground, there is a paved area with some benches and a few small trees. The overall scene is a blend of modern architecture and natural beauty.

Edgar Lobaton, Ph.D.

Director of Applied AI, College of Engineering (COE), NC State
Professor, Electrical and Computer (ECE) Engineering, NC State
NAIRR AI Education Fellow, Computing Research Association (CRA)

The Problem: Synthetic Gait Classification

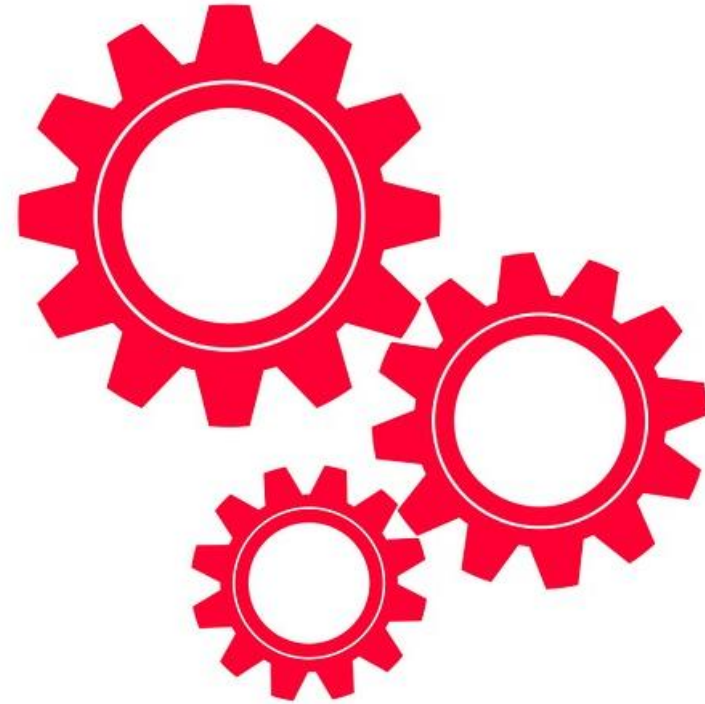
- We are aiming to develop a personalized model that determine when a person is walking fast or slow. We have conducted two recording sessions, in which the individual switches between fast and slow walking periodically.
- Data: <https://drive.google.com/drive/folders/19RpNHQ0FrzTCWYapPC1PsTkQc2tL4979?usp=sharing>

The Problem: Synthetic Gait Classification

- Let's do this as a Self-Automator

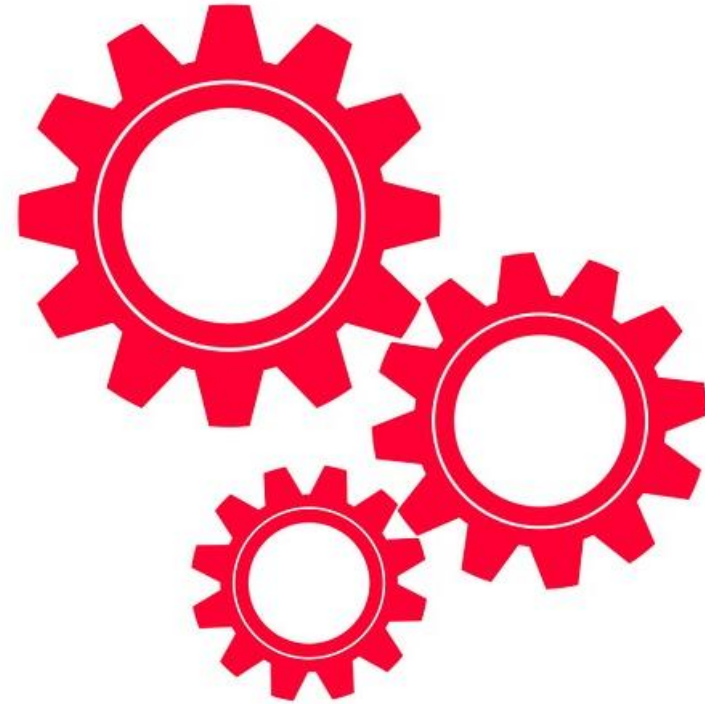
On the Hub:

- Open a terminal and run
 - `mkdir Gait-Automator`
 - `Cd Gait-Automator`
- Upload the files
- Open cline



The Problem: Synthetic Gait Classification

- Let's do this as a Centaur



The Problem: Synthetic Gait Classification

- Let's refine as a Cyborg

